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Collation

Collation is the assembly of written information into a standard order. Many systems of collation are based on numerical order or alphabetical order, or extensions and combinations thereof. Collation is a fundamental element of most office filing systems, library catalogs, and reference books.

Collation differs from *classification* in that classification is concerned with arranging information into logical categories, while collation is concerned with the ordering of items of information, usually based on the form of their identifiers. Formally speaking, a collation method typically defines a total order on a set of possible identifiers, called sort keys, which consequently produces a total preorder on the set of items of information (items with the same identifier are not placed in any defined order).

A collation algorithm such as the Unicode collation algorithm defines an order through the process of comparing two given character strings and deciding which should come before the other. When an order has been defined in this way, a *sorting algorithm* can be used to put a list of any number of items into that order.

The main advantage of collation is that it makes it fast and easy for a user to find an element in the list, or to confirm that it is absent from the list. In automatic systems this can be done using a binary search algorithm or interpolation search; manual searching may be performed using a roughly similar procedure, though this will often be done unconsciously. Other advantages are that one can easily find the first or last elements on the list (most likely to be useful in the case of numerically sorted data), or elements in a given range (useful again in the case of numerical data, and also with alphabetically ordered data when one may be sure of only the first few letters of the sought item or items).

Contents

Numerical and chronological order

Strings representing numbers may be sorted based on the values of the numbers that they represent. For example, "−4", "2.5", "10", "89", "30,000". Note that pure application of this method may provide only a partial ordering on the strings, since different strings can represent the same number (as with "2" and "2.0" or, when scientific notation is used, "2e3" and "2000").